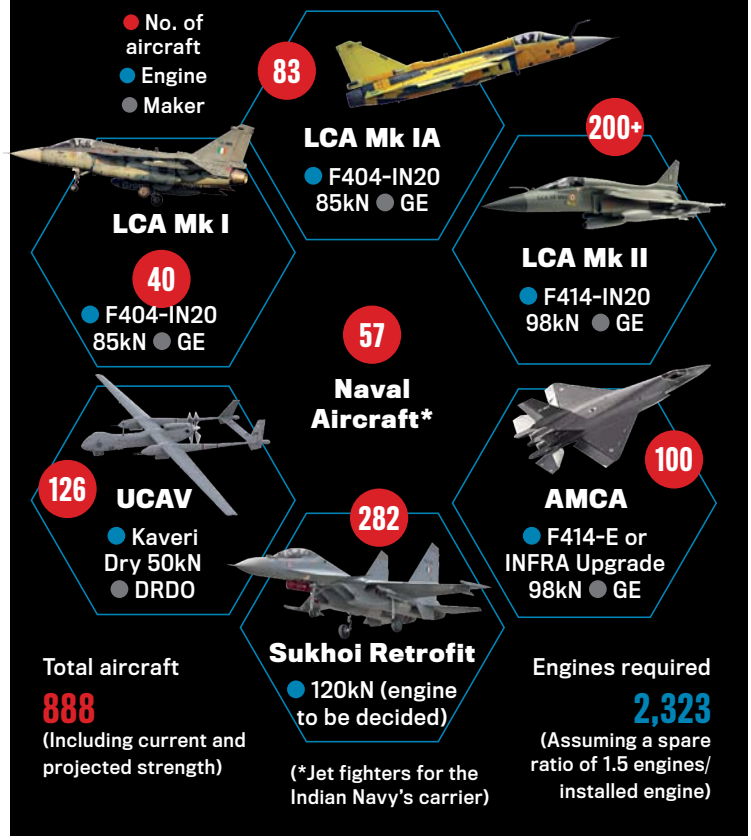


TURBO CHARGED

The Indian Air Force and Indian Navy's combat jet expansion plans would require over 2,300 engines



Graphic by TANMOY CHAKRABORTY

thrust of about 75kN (kilonewtons).

The Indian Air Force (IAF) has opened its hunt for 114 multi-role fighters while the Navy needs approximately 30 carrier-based fighter jets. Their combined value? Some $20 billion. The IAF also requires close to 600 India-made fighters for its fleet, all of which would require over 2,300 engines, assuming a spare ratio of 1.5 engines/ installed engine. Moreover, the Sukhoi fleet of 282 jets will also go in for engine retrofit in the years to come. Importing all these engines will involve a significant spend of India's foreign exchange.

This immense business opportunity has global aerospace players making a beeline for India with jet engine technology, which has been closely guarded so far. And the fiercest contest is between the US and France. Foreign players are willing to sell or indigenise their products to meet Indian requirements, but their offers are conditional on large orders.

India has proved its mettle in developing nuclear submarines or aircraft carriers or even fighter jets, but any progress in combat jet engine development has remained elusive. India's quest for combat jet engines has been fraught since the days of the first indigenous jet fighter, the HF-24 Marut in the 1960s. The same issue has dogged the Tejas LCA (light combat aircraft) programme and the ongoing development of a fifth-generation Advanced Medium Combat Aircraft (AMCA).

Very few countries—the US, France, the UK and Russia—have succeeded in mastering the complex technologies needed to produce engines for combat aircraft. China is still using Russian engines for its J-20 fighters due to the lack of required thrust in its own WS-10 engines that were developed with a massive budget. Since 1986, India has developed nine prototypes of its Kaveri engine, which have failed to meet the required parameters to power a fighter. The Defence Research and Development Organisation's 'Gas Turbine Research Establishment (GTRE), tasked to develop the Kaveri engine, has been unable to deliver one that could power the LCA despite a cost overrun of 642 per cent and a delay of about 13 years', the Comptroller and Auditor General noted in its report released in 2011.

India's mainstay LCA Tejas flies on American General Electric (GE) F404 engines, and the Tejas Mk2 and AMCA, now under development, will be powered by GE's F414 engines. IAF would require nothing less than 170 Tejas Mk2, which is equivalent to six squadrons. In addition, it has also agreed to induct seven squadrons of AMCA. GE is expected to set up a plant in India to manufacture F414 engines in collaboration with HAL and other private players. The company has agreed to transfer manufacturing technology and production engines but is awaiting the US government's nod. The Super Hornet is powered by two GE F414-400 afterburning turbofan engines while the French Rafale-M will have Safran's M88 engine. But India's military planners believe if the country is looking at jets for 20 years, it needs to take an immediate call on engine manufacturing.

"You cannot design an aircraft and

IN THE FRAY

Five companies are thriving in jet engine design. The US, UK and France have offered to join hands with India

USA

GE Aviation

Pratt & Whitney

UK

Rolls-Royce

FRANCE
Safran

RUSSIA
NPO Saturn